



Department of Computer Science and Engineering (Data Science)

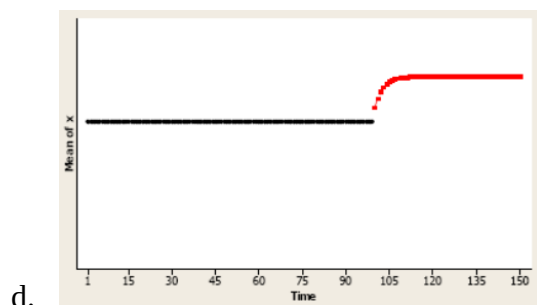
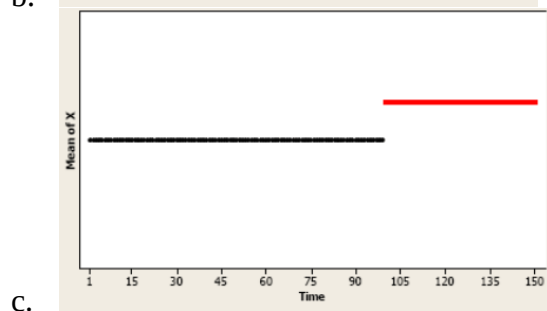
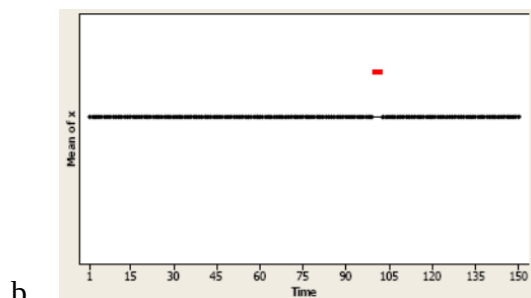
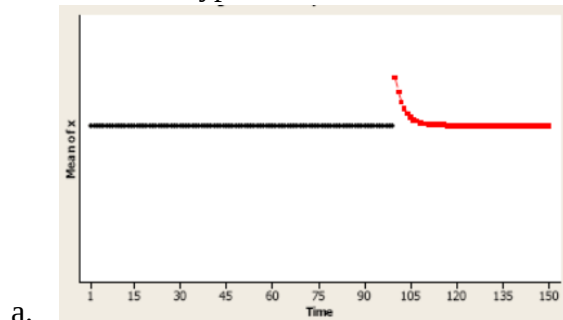
Practice Questions – TT2

1. Explain in detail the ADF test with respect to unit roots.
2. Explain in detail the different types of detrending techniques used to detrend a time series.
3. Determine from the below observations if the time series data is stationary or not using then ADF test and also justify with respect to the conditions.
Test Statistic: 0.815369
p-value: 0.991880
Critical Values:
5%: -2.884
1%: -3.482
10%: -2.579
4. Determine from the below observations if the time series data is stationary or not using then ADF test and also justify with respect to the conditions.
Test Statistic: -4.444805
p-value: 0.000247
Critical Values:
5%: -2.884
1%: -3.482
10%: -2.579
5. Determine from the below observation and complete wherever necessary and find:
 - a. If the time series data contains unit roots or no.
 - b. Complete the missing values.
 - c. Mention the statistical test used to check the presence of unit roots.
 - d. Justify the conclusion.
Test Statistic: -4.444805
p-value: 0.000247
Critical Values:
5%: -----
1%: -----
10%: -----
6. Explain and evaluate the following:
 - a. Importance of Residual diagnostic.
 - b. With the help of an example explain the Naïve forecast method to calculate residuals.
7. A researcher is testing whether a financial time series has a unit root using the ADF test. However, the distribution of the test statistic is unknown. Explain how the Monte



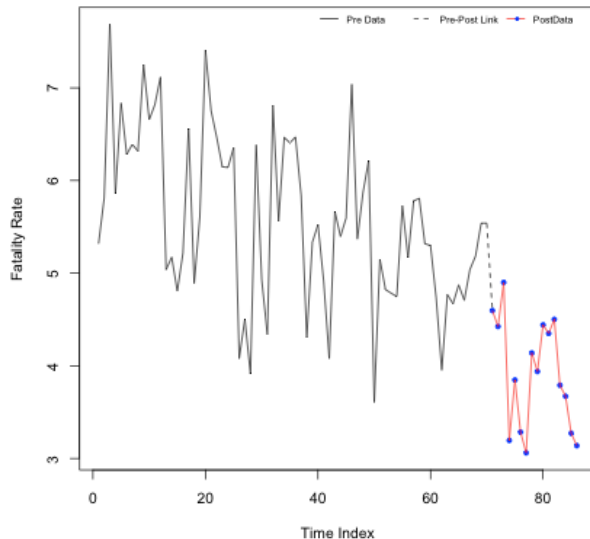
Carlo method can be applied to generate critical values for the Dickey-Fuller test in this scenario. Illustrate the process with steps.

8. Explain intervention analysis with respect to the ARIMA model.
9. Sketch the various patterns of intervention analysis effect using suitable graphs.
10. Explain how Monte Carlo simulate and analyze the behaviour of a given time series model.
11. Differentiate the type of intervention effect based on the graphs given.





12. North Carolina Highway Fatality Rates, monthly for $n = 86$ months. Beginning at month = 71 the maximum speed limit was lowered from 70 mph to 55 mph. The fatality rate is computed as highway fatalities per 100 million miles of travel. A time series plot follows. The latter part of the series, in red, is the fatality rate after the change in speed limit. Determine the type of method/effect used to determine the fatality rate.

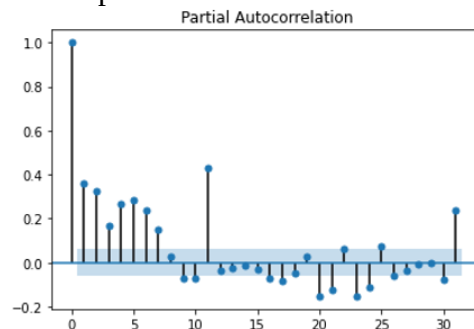


13. Explain and derive the ADL multivariate model with respect to transfer function for any order p, q where $p=2$ and $q=2$.
14. Explain in detail the need and the steps for Monte-Carlo Method.
15. You are given two-time series datasets—one for stock prices and another for exchange rates. How would you apply a VAR model to analyze the interdependence between the two series and assess the effect of exchange rate fluctuations on stock prices over time?
16. An economist observes that the relationship between GDP growth and inflation changes significantly before and after a financial crisis. Explain how a TVP-VAR model can be applied to capture this change and what insights it provides compared to a standard VAR.
17. A data scientist is building a forecasting model for energy demand using multiple variables like temperature, industrial output, and population, but the model becomes too complex and gives unstable predictions. Explain how BVAR can be applied to improve the model and describe the role of priors.
18. Which statistical model is used to analyze volatility in time series in order to forecast future volatility. Also, which the help of the model constructs a (p) model for the same.
19. The Nigerian Stock Exchange has grown tremendously over the years in an attempt to meet the ever increasing needs and sophistication of the domestic economy; it also plays a significant role in the daily capital formation and resource generation of the



nation. The Nigerian Stock Exchange as an essential place for sourcing funds needed to power the financial needs of industries is highly dependent on the steadiness (stability) of prices of stock on the exchange. There is a major unsteadiness (instability) in stock prices. The company wants to predict the returns of the stock price based on the past values and the instability.

- State the appropriate model to be used for the scenario above and justify.
- Based on the scenario selected state the order of the model referring to the below PACF plot.



- Write the model with respect to the order selected in the question above.

20. Which of the following statements are true concerning the standardised residuals (residuals divided by their respective conditional standard deviations) from an estimated GARCH model?

- They are assumed to be normally distributed.
- Their squares will be related to their lagged squared values if the GARCH model is appropriate
- In practice, they are likely to not work with volatility.
- If the GARCH model is only considering the past errors

21. Explain the ARCH model with respect to the below points:

- Each component in ARCH model.
- Need of ARCH model.
- When to apply an ARCH model.
- ARCH(p).

22. Which of the following is TRUE about ARCH and GARCH models?

- They are used for modelling and forecasting volatility
- They are non-linear models
- They can both be estimated using OLS(IV) Series estimated using these models must have a unit root process.



23. Explain the importance of ARCH model and derive the equation for ARCH (2) and ARCH (p) model.
24. You are the manager of a supermarket and would like to forecast the sales in the next few weeks and have been provided with the historical daily sales data of hundreds of products. But, as instructed by the CEO, you have been asked to change the price of some products as the manufacturing cost has increased. The management has asked to submit a detailed analysis between the past price value of the product and the changed value to check if it has hampered the total sales of the product for their supermarket.
- k. State the model that will be used to analyse the data.
 - l. Apply the model used above with the help of various notations.
 - m. State the various ways in which the data can be seen on the analysis submitted to the management.
25. Differentiate between ARCH and GARCH model based on the notations used to justify the model.
26. Differentiate between SV and Bilinear VAR.
27. A company wants to develop a predictive model quickly. They are considering AutoML instead of a traditional ML workflow. Apply your knowledge to evaluate both approaches and recommend when AutoML should not be used.
28. Given a dataset with multiple time-dependent variables, explain how multivariate time series regression can be applied. How does it differ from univariate regression, and in which scenarios is it useful?
29. Explain convolution with respect to time domain and prove that the convolution in time domain is equivalent to the multiplication of their spectrum in frequency domain.
30. State the problem of convolution in time domain with respect to the points in the convolution result. Also, with the help of the formula provide a solution depicting a pictorial representation.
31. During festivals, the promotion of barbecue meat will also boost the sales of ketchup and other spices. If one brand of toothpaste is on sale, the demand of other brands might decline. Which type of analysis needs to be done when various variables(products) forecast is dependent on some other variable (product)?
32. Apply an appropriate model based on the following parameters given:



ARCH (3)
Intervention Model
ADL (4,4)
GARCH(3,2)

33. A time series shows nonlinear dependence where past values interact with past errors (e.g., demand influenced by past demand and shocks). Explain how a Bilinear AR model can be applied to model this behavior and why linear AR models fail.

34. A model is defined as:

$$y_t = 0.5y_{t-1} + 0.2y_{t-2} + 0.3x_{t-1} + 0.1x_{t-2} + \epsilon_t$$

Given:

$$y_{t-1} = 2 ; y_{t-2} = 1 ; x_{t-1} = 3 ; x_{t-2} = 2$$

- Explain the role of lag length in a VAR model.
- Determine whether this represents a complete VAR system. Justify.
- Compute y_t .

35. A system is designed to predict future electricity load using only its past values.

- Identify the appropriate nonlinear model for this scenario.
- Write the general form of the model.
- Explain how this model works in this context.

36. A dataset shows non-stationary behavior and the analyst wants to automatically build a forecasting model without manually selecting parameters.

- Explain how Auto-ARIMA can be used in this situation.
- Describe the steps it will follow to select the best model.

37. In an AutoML pipeline, both hyperparameter tuning and ensemble methods are used. Analyze their roles and explain how they contribute differently to improving model accuracy.

38. A dataset contains millions of unique store and product IDs, making traditional encoding techniques impractical.

- Explain why one-hot encoding is not suitable for such high-cardinality categorical data.
- Describe how feature hashing can be applied in this scenario to handle the data efficiently.
- Discuss the role of the hash size K and explain the trade-off between dimensionality reduction and model accuracy.